Active Control of a Laser-Levitated Photonic Metasurface

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Project Overview

Goal: Develop an active control scheme to stabilize and move a photonic metasurface which is levitated by a collimated laser beam.

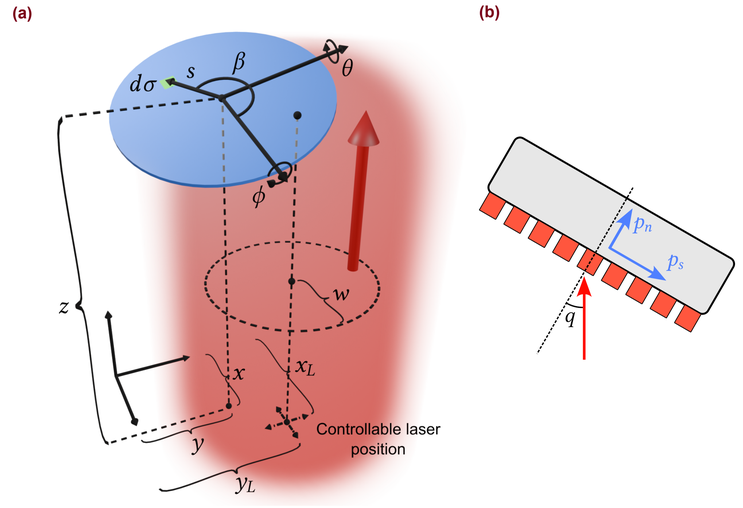


Figure 1. A collimated beam incident at angle q on the metasurface induces normal and radial radiation pressure components $p_n(q,s)$ and $p_s(q,s)$ which govern dynamics of the surface. The laser position in the x-y plane and (optionally) its intensity are used to control the surface position and orientation.

Metasurface Dynamics

We may calculate the force and torque on the metasurface by integrating on its surface. The normalized beam profile $I(\mathbf{r}_\perp)$ is given as a function of the distance from the beam axis $\mathbf{r}_\perp$, and scaled by max intensity I_0.

$$\mathbf{F}_{\text{body}} = I_0 \int_0^{2\pi} d\beta \int_0^{D/2} s\, ds \begin{bmatrix} p_s(s,q)\cos\beta \\ p_s(s,q)\sin\beta \\ p_n(s,q) \end{bmatrix} I(\mathbf{r}_\perp)\cos\theta\cos\phi \quad (1)$$

$$\boldsymbol{\tau} = I_0 \int_0^{2\pi} d\beta \int_0^{D/2} s\, ds \begin{bmatrix} s\cos\beta \\ s\sin\beta \\ 0 \end{bmatrix} \times \begin{bmatrix} p_s(s,q)\cos\beta \\ p_s(s,q)\sin\beta \\ p_n(s,q) \end{bmatrix} I(\mathbf{r}_\perp)\cos\theta\cos\phi \quad (2)$$

The force vector is transformed from the surface body frame B to the fixed frame I via a rotation matrix $H_B^I(\phi,\theta,\psi)$.

Active Control Scheme

Dynamics cast as a first order ODE in the state and control vectors which may then be linearized about a fixed point,

$$\mathbf{x} = \begin{bmatrix} x\ y\ z\ \phi\ \theta\ \psi\ v_x\ v_y\ v_z\ \omega_1\ \omega_2\ \omega_3 \end{bmatrix}^T, \quad \mathbf{u} = \begin{bmatrix} x_L\ y_L\ I_0 \end{bmatrix}^T \quad (3)$$

$$\dot{\mathbf{x}} = f(\mathbf{x},\mathbf{u}) \rightarrow \dot{\mathbf{x}} \approx A\Delta\mathbf{x} + B\Delta\mathbf{u} \quad (4)$$

Here $A = \left.\frac{df}{d\mathbf{x}}\right|_{\mathbf{x}^*,\mathbf{u}^*}$ and $B = \left.\frac{df}{d\mathbf{u}}\right|_{\mathbf{x}^*,\mathbf{u}^*}$ with $\mathbf{x}^*$ and $\mathbf{u}^*$ a fixed point wherein the metasurface is flat and unmoving, and the beam is centered on the metasurface. Typical observations of the system only include position,

$$\mathbf{y} = C\mathbf{x} = \begin{bmatrix} x\ y\ z \end{bmatrix}^T \quad (5)$$

A Kalman filter gives a dynamical equation to update the state estimate $\hat{\mathbf{x}}$, which we use to update our controller $\mathbf{u} = -K\hat{\mathbf{x}}$.

Simulation of Specular Reflector & Control Scheme

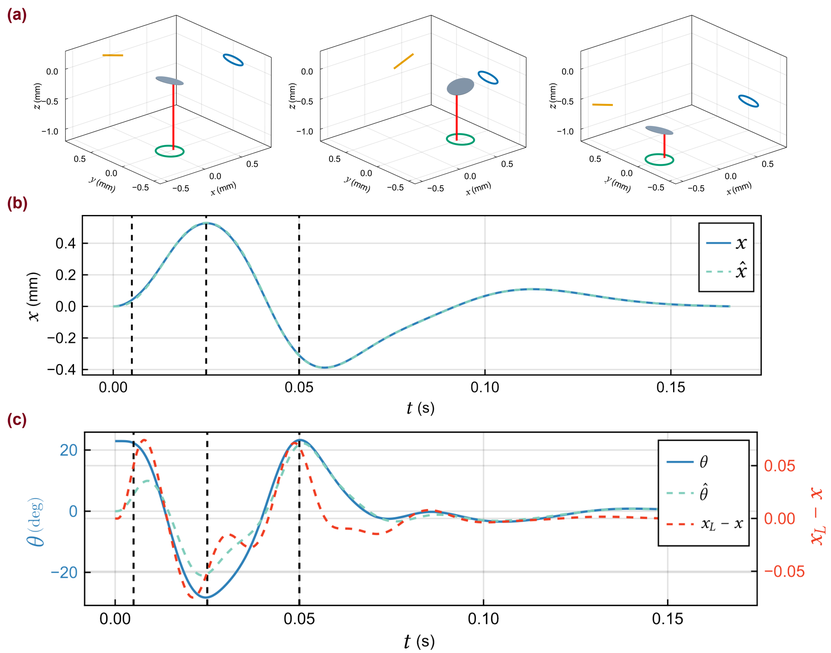


Figure 2. Trajectory of specular reflecting disc with $\theta_0 = 23°$. a) Snapshots of configuration, b), c) x and θ coordinates as a function of time. Vertical dashed lines are snapshot times.

Adapting Control Scheme to Experiment

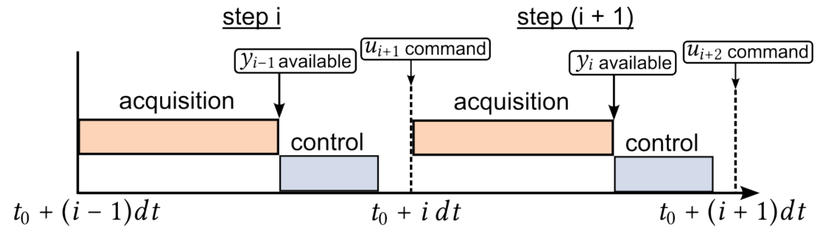


Figure 3. Experimental control loop with data acquisition (e.g. imaging the metasurface, analyzing for position) and control steps.

Control loop for step i:

1. Use $\mathbf{y}_{i-1}$ and $\mathbf{u}_{i-1}$ to update $\hat{\mathbf{x}}_{i-2} \rightarrow \hat{\mathbf{x}}_{i-1}$.
2. Use the linear dynamics A, B and $\mathbf{u}_i$ to project forward $\hat{\mathbf{x}}_{i-1} \rightarrow \overline{\mathbf{x}}_i$.
3. Use to projected state vector to calculate control value $\mathbf{u}_{i+1} = -K\overline{\mathbf{x}}_i$.

All matrix operations, and may be done in real time.

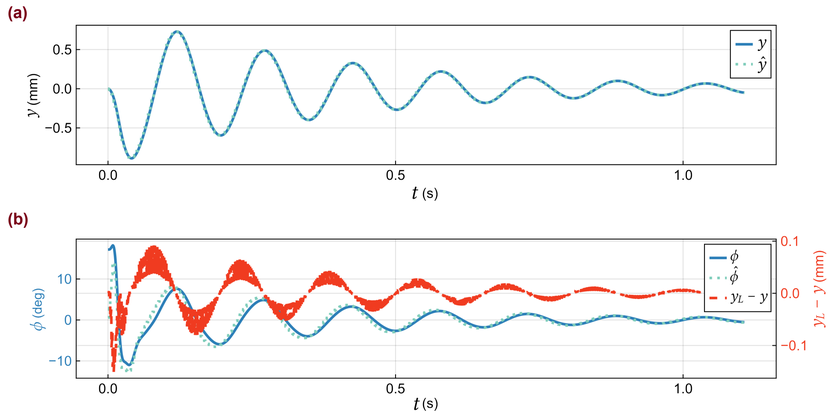


Figure 4. Metasurface trajectory with $\phi_0 = 17.2°$ and control delay 2ms.

References

"Optomechanical Self-Stability of Freestanding Photonic Metasurfaces" Kumar, Kindem, Ilic *Phys. Rev. Applied*, 16, 014053 (2021).

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